



Muscle fiber type diversification during exercise and regeneration

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ABSTRACT

The plasticity of skeletal muscle can be traced down to extensive metabolic, structural and molecular remodeling at the single fiber level. Skeletal muscle is comprised of different fiber types that are the basis of muscle plasticity in response to various functional demands. Resistance and endurance exercises are two external stimuli that differ in their duration and intensity of contraction and elicit markedly different responses in muscles adaptation. Further, eccentric contractions that are associated with exercise-induced injuries, elicit varied muscle adaptation and regenerative responses. Most adaptive changes are fiber type-specific and are highly influenced by diverse structural, metabolic and functional characteristics of individual fiber types. Regulation of signaling pathways by reactive oxygen species (ROS) and oxidative stress also plays an important role in muscle fiber adaptation during exercise. This review focuses on cellular and molecular responses that regulate the adaptation of skeletal muscle to exercise and exercise-related injuries.

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1. Introduction

Skeletal muscle is a highly dynamic tissue that undergoes continuous remodeling in response to various metabolic and functional demands. The quantity and quality of muscle can be traced down to the structural and contractile proteins that respond to physiological and pathological conditions including exercise and injury [1]. Individual muscle fibers vary in their mechanical, biochemical and metabolic properties depending upon the fiber type. Various criteria have been used to classify fiber types including histochemical methods [2,3], speed of twitch contraction [4], fatigability, dominant enzymatic pathway and the

Abbreviations: ATF, Activating Transcription Factor; CaMK, Ca^{2+} /Calmodulin-dependent protein kinase; CAT, Catalase; CREB, cAMP Response Element-binding Protein; EE, endurance exercise; ER α , Estrogen Receptor α ; FOXO, Fork Head Box; GLUT-4, Glucose Transporter-4; GPX, glutathione peroxidase; GR, Glucocorticoids Receptor; HIF-1- α , hypoxia inducible factor 1- α ; HNF4, Hepatocytes Nuclear Factor-4; IGF-1, Insulin like Growth Factor-1; LXR, Liver X Receptor; MAPK, Mitogen Activating Protein Kinase; MEF-2, Myocytes Enhance Factor-2; MND, myonuclear domain; LDH, Lactate Dehydrogenase; MRF, myogenic regulatory factor; MyHC, myosin heavy chain; NFAT, nuclear factor of activated T-cells; NF κ B, nuclear factor κ B; NMJ, Neuromuscular Junction; NRF, Nuclear Factor Erythroid-2 related factor; PFK, Phosphofructo Kinase; PGC-1, proliferator-activated receptor- γ coactivator-1 α ; PPAR α , Peroxisome Proliferator-activated Receptor α ; RE, resistant exercise; Rheb, Ras homolog enriched in brain; RNS, reactive nitrogen species; ROS, reactive oxygen species; SDH, Succinate Dehydrogenase; SOD, superoxide dismutase; VEGF, Vascular Endothelial Growth Factor; TR, Thyroid Receptor

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myosin heavy chain (MyHC) isoform expression [5]. Of those, MyHC isoforms are the most frequently used classification criteria and are considered the molecular markers of fiber types. Myosin is the molecular motor and the prime driving protein in force generation. It is also the most abundant protein in the sarcomere comprising $\approx 25\%$ of the total muscle proteins [6]. Due to its abundance and contractile significance, qualitative and quantitative changes in myosin and its isoforms have significant effects on muscle strength. Human limb muscles contain three isoforms of MyHC called type I, type IIa and type IIx, and a fiber can express a single MyHC isoform (i.e., pure fiber) or co-express multiple isoforms (i.e., hybrid fiber) [7]. Rodent muscles additionally contain type IIb and IIx fibers [8]. Type I fibers are called slow-twitch fibers because of their slow speed of contraction. They have a predominantly oxidative metabolism. Type IIb and IIx fibers are fast-twitch fibers because of their fast speed of contraction. They mainly metabolize glucose by glycolytic pathway. Type IIa fibers are intermediate fibers with fast speed of contraction but mixed (glycolytic/oxidative) metabolism [9]. The frequency of hybrid fibers increases under various stimuli and relates to high degree of muscle plasticity, with exercise and disuse being prime determinants of muscle fiber type transition [10,11]. Because fiber type diversity is associated with functional diversity, alterations in muscle fiber types affect contractile, metabolic and biochemical properties of the muscle.

Exercise training is one of the prime modulators of muscle plasticity as it triggers a series of intracellular signaling pathways which mediate muscle growth and adaptation [12]. These

adaptations include changes in contractile proteins structure and function [13], satellite cells and myonuclei [14], mitochondrial homeostasis [15], metabolic profile [16] and muscle capillary density [17]. Different modalities of exercise are possible depending upon the type, intensity and duration of contraction.

Contractile activity generates a complex set of reactive oxygen species (ROS) and reactive nitrogen species (RNS) in the skeletal muscle. These reactive species can positively or negatively modulate muscle force generation depending upon the concentration and temporal pattern on ROS generation [18], and play important role in contraction-induced muscle adaptation [19]. For the purpose of this review we will mainly focus on endurance (aerobic) exercise (EE) and resistance (strength) exercise (RE) which have become central issues in sports science and clinical settings. While these two types of exercises have many combined health benefits, they have been studied distinctly because of their divergent effects on various muscle parameters discussed later in this review. We will review the impact of muscle fiber type heterogeneity on exercise performance. We will also discuss various types of exercise affecting muscle fiber type diversity and the role of ROS production in this process. In the final section of this review we will highlight exercise-induced injuries and regeneration potential of skeletal muscle and individual fiber types.

2. Muscle fiber type diversification and exercise performance

2.1. Muscle fiber types

There are four major fiber types found in the skeletal muscles of limb and trunk in various proportions. The relative proportion of the fiber types in a given muscle may vary according to specie and the functional assignment of the muscle [8,20]. The diverse population of the muscle fibers in a given muscle allows for various types of tasks ranging from prolonged, low intensity contractions (e.g., to maintain posture) to fast and strong maximal contractions (e.g., kicking). This tremendous range of tasks is attributed to diversity in functional cell compartments in individual muscle fibers including membrane excitability, calcium transients, energy supply systems and the contractile machinery in the sarcomere (Fig. 1).

The diversity of muscle fibers allows them to perform specialized functional tasks. Thus the type I fibers with high oxidative capacity and capillary density are more suitable for endurance exercise [21] while type IIb fibers with low oxidative capacity and capillary density are suitable for short term RE [22]. Type IIa are the intermediate fiber types that allow high power generation at a considerable velocity with good endurance.

2.2. Muscle fiber type plasticity

The fiber type composition of a muscle, once thought to be genetically determined [23], is highly plastic and can be altered in response to functional demands including neuromuscular stimulation [24], mechanical loading [25], hormones [26] and aging [27]. Exercise induced changes in fiber type transition are determined by frequent nerve stimulation resulting in an increased duration of elevated cytosolic free Ca^{2+} [28]. It is believed that calcineurin, a calcium regulated serine/threonine phosphatase plays central role in fiber type specific gene regulation. Indeed, selective up regulation of calcineurin promotes type I fibers gene products while inhibition of calcineurin promotes type II fibers-specific gene activity [29]. This fiber type switching is controlled via calcineurin mediated activation of nuclear factor of activated T-cells (NFAT), which are a family of transcription factors involved in nerve activity sensing and calcium regulation [30]. A number of other

Morphology						
Fiber type	Cross-Sectional area	Capillary density	Satellite Cell count	NMJ size	Nuclei count	MND size
Type I Oxidative	▲	▲	▲	▲	▲	▲
Type II Glycolytic	▲	▲	▲	▲	▲	▲
Contractile Properties						
Fiber type	Force Generation	Contraction Velocity	Time to Peak Tension	$\text{Ca}^{2+}/\text{Mg}^{2+}$ ATPase activity	Endurance Capacity	Fatigue Resistance
Type I Oxidative	▲	▲	▲	▲	▲	▲
Type II Glycolytic	▲	▲	▲	▲	▲	▲
Bioenergetics Properties						
Fiber type	Mitochondria Density	Glycogen Phosphorylase	PFK activity	SDH activity	LDH activity	Citrate Synthase
Type I Oxidative	▲	▲	▲	▲	▲	▲
Type II Glycolytic	▲	▲	▲	▲	▲	▲
Protein Dynamics						
Fiber type	Protein turnover	mRNA content	Half life of Myosin	IGF-1 expression	Myostatin expression	MAFbx/MuRF expression
Type I Oxidative	▲	▲	▲	▲	▲	▲
Type II Glycolytic	▲	▲	▲	▲	▲	▲

Fig. 1. Characteristics of individual fiber types in mammalian skeletal muscles. NMJ (Neuromuscular Junction); MND (Myonuclear Domain); PFK (Phosphofructo Kinase); SDH (Succinate Dehydrogenase); LDH (Lactate Dehydrogenase); IGF-1 (Insulin like Growth Factor-1).

transcription factors, co-activators and co-repressors have also been associated with fiber type switching and have been comprehensively reviewed elsewhere [31].

2.3. ROS generation by muscle fibers

Marked differences exist between fast and slow twitch fibers with regard to production and metabolism of reactive oxygen species. Amplex red measurements of H_2O_2 from permeabilized muscle fibers [32] and isolated mitochondria [33] show up to three fold higher H_2O_2 release in fast-twitch gastrocnemius compared to slow-twitch soleus muscle in the presence of complex I or complex II substrate. This difference is attributed to fiber type-specific variations in endogenous H_2O_2 scavenging capacities. Indeed, direct measurements from isolated permeabilized fibers show that the mitochondria from slow-twitch fibers have an approximately two fold higher H_2O_2 scavenging capacity compared to mitochondria from fast-twitch fibers [34]. Further, the slow-twitch fibers also show higher activities of anti-oxidant enzymes superoxide dismutase (SOD) [35], glutathione peroxidase (GPX) [36] and Catalase (CAT) [37] compared to fast-twitch fibers. These differences are attributed to differential expression of proliferator-activated receptor- γ coactivator-1 α , PGC-1 α and PGC-1 β . This transcription coactivator, apart from inducing mitochondrial biogenesis also regulates the expression level of key antioxidant enzymes described above [38]. Taken together, these data highlight striking differences between H_2O_2 emitting capacity and buffering potential between two fiber types.

3. Endurance exercise

Endurance exercise (EE) is characterized by repeated, sustained, low intensity contractions for a prolonged period of time without getting fatigued. The term generally refers to training the aerobic system (Krebs cycle, oxidative phosphorylation) versus the anaerobic system. The force production is relatively small ($\approx 30\%$) relative to maximum force-generating capacity of the muscle [39]. Examples of this type of exercise include long distance running, cycling and swimming. EE involves maintaining a given power output for the longest possible time and is characterized by numerous physiological benefits (Table 1).

Endurance training results in changes in muscle fiber type composition which are mainly restricted to type II fibers and involve a transformation from IIb to IIa fibers resulting in a more oxidative muscle [40]. Type I fibers are more economical in energy utilization during the cross-bridge cycle (i.e., lower adenosine triphosphatase activity) compared to type II fibers, and are hence important in the energy efficiency of the muscle. The response in contractile properties of individual fibers is mainly limited to contractile speed where fast-twitch fibers become slower and slow-twitch fibers become faster. These changes in contractile speed are attributed to a shift in myosin light chain isoforms from fast to slow and slow to fast type, for the two isoforms respectively [40,41].

3.1. EE enhances mitochondrial biogenesis

A hallmark response of endurance training is mitochondrial biogenesis [42] coupled with improved functional parameters of mitochondria [43]. These responses might be linked to changes in concentrations of cellular metabolites which occur after chronic stimulation of skeletal muscle [44]. For instance, 6 weeks of exercise training increases the muscle mitochondrial content by ≈ 50 –100% [45]. An increase in number and volume of mitochondria is attributed to but not limited to increased PGC-1 α expression [46–48], the master regulator of mitochondrial biogenesis (Fig. 2). This is further supported by blunting of exercise-induced mitochondrial biogenesis in mice deficient for PGC-1 α [49]. PGC-1 α has four different isoforms (PGC-1 α 1, PGC-1 α 2, PGC-1 α 3, PGC-1 α 4) with alternate promoter usage and splicing of the primary transcript [50, 51]. Among them, PGC-1 α 1 which transcribes proximal promoter of PGC-1 α gene, gets activated in response to EE and is mainly involved in oxidative phenotype (Fig. 2).

This phenotype is achieved via a group of nuclear and mitochondrial transcriptional factors summarized in Table 2.

To our knowledge the exercise-induced role(s) of PGC-1 α

Table 1

Skeletal muscle adaptive response to endurance and resistant exercises. $\leftrightarrow$, No change; $\leftrightarrow\uparrow$ no change or small effect; $\uparrow$, small effect; $\uparrow\uparrow$, large effect.

	Endurance exercise	Resistance exercise
Muscle hypertrophy	$\leftrightarrow$	$\uparrow\uparrow$
Muscle strength	$\leftrightarrow$	$\uparrow\uparrow$
Muscle fiber size	$\leftrightarrow\uparrow$	$\uparrow\uparrow$
Myofibrillar protein synthesis	$\uparrow$	$\uparrow\uparrow$
Satellite cells count	$\uparrow$	$\uparrow\uparrow$
Myonuclei count	$\leftrightarrow\uparrow$	$\uparrow\uparrow$
Lactate tolerance	$\uparrow\uparrow$	$\leftrightarrow\uparrow$
Glycolytic function	$\uparrow$	$\uparrow\uparrow$
Mitochondrial volume	$\uparrow\uparrow$	$\uparrow$
Mitochondrial protein synthesis	$\uparrow\uparrow$	$\leftrightarrow\uparrow$
Capillary density	$\uparrow\uparrow$	$\leftrightarrow$
Oxidative function	$\uparrow\uparrow$	$\leftrightarrow\uparrow$
Endurance capacity	$\uparrow\uparrow$	$\leftrightarrow\uparrow$

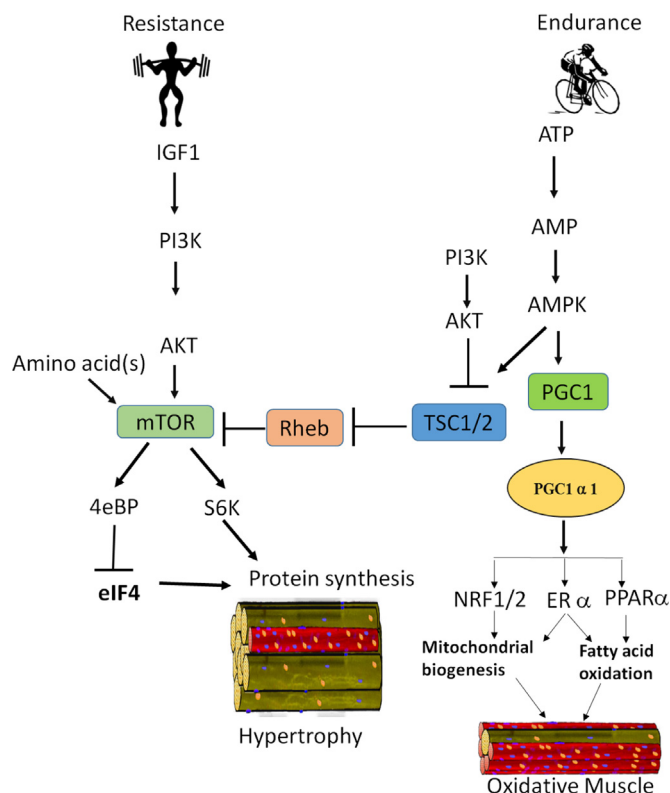


Fig. 2. Intracellular signaling pathways regulate different muscle phenotypes in response to resistant and endurance exercises.

Table 2

List of transcription factors interacting with PGC-1 α to regulate genes involved in muscle adaptation process to exercise. NRF (Nuclear Respiratory Factor); PPAR (Peroxisome Proliferator-Activated Receptor); ERR (Estrogen Related Receptor); TR (Thyroid Receptor); LXR (Liver X Receptor); MEF2 (Myocytes Enhancer Factor 2); FOXO (Fork Head Box); ATF2 (Activating Transcription Factor 2); CREB (cAMP Response Element Binding Protein).

Transcription factors/nuclear receptors	Function	Reference
NRF1	Mitochondrial respiratory capacity and glucose transport	[52]
NRF2	Muscle regeneration and metabolic regulation	[53]
PPAR α , β/δ	Fatty acid oxidation	[54]
ERR α , β/δ	Homeostasis and regeneration	[55]
TR β	Homeostasis and repair	[56]
LXR α , β	Fatty acid oxidation	[57]
MEF2	Homeostasis and repair	[58]
FOXO1	Homeostasis and mitochondrial metabolism	[59]
ATF2	Metabolic adaptation	[60]
CREB	Muscle regeneration and homeostasis	[61]

isoforms are not studied in a fiber type-specific manner.

Among the human muscle fiber types, the mitochondrial content is highest in type I fibers followed by type IIa and type IIx fibers (Fig. 1) [62,63]. All three fiber types show an increase in PGC-1 α expression after 6 weeks of exercise [21,64]. The response is intensity-specific whereas lower intensity exercise ($\approx 50\%$ $\text{VO}_{2\text{max}}$) predominantly increase the mitochondrial volume in type I fibers while much higher intensity exercise is required to recruit mitochondria in type II fibers [21,65]. These changes might delay the recruitment of type II fibers during the exercise helping the muscle energy efficiency in a positive manner [66]. In addition to

mitochondrial content, the dynamic regulation of fission and fusion machinery is also considered integral part of muscle metabolic adaptation. Mitochondrial fusion is mainly controlled by mitofusion 1 and 2 (MFN 1 and MFN 2) in the outer mitochondrial membrane and optic atrophy type 1 (OPA 1) in the inner mitochondrial membrane. Fission is mainly controlled by dynamin-related protein 1 (Drp1) and its recruitment factors fission 1 (Fis1) and mitochondrial fission factor (MFF) in the outer membrane [67]. Chronic exercise triggers the expression of mitofusion 1 and 2 [68,69] which leads to increased mitochondrial volume. It is postulated that the elongated mitochondrial networks facilitate rapid transmission of membrane potential across greater distances in the cell [70]. Further, the fission machinery responds to EE by increasing the expression of Drp1 and Fis1 [68,71,72] although this effect is blunted in the presence of insulin resistance [73]. Hence, fission might be required to remodel the reticulum during proliferation and/or re-locate parts of it via fragmentation [70].

In addition to mitochondrial biogenesis and remodeling, EE-induced muscle adaptation likely requires effective removal of damaged and/or dysfunctional mitochondria. Mitophagy is an evolutionary conserved process for lysosome-dependent degradation of mitochondria. The regulatory effects of EE on skeletal muscle mitophagy are only beginning to emerge. EE along with mild caloric restriction prevents the age-related reduction in basal autophagy in the rat skeletal muscle [74] which is a beneficial effect as deficiency in basal autophagy is associated with muscle damage [75]. Recently, EE was shown to induce the expression of mitophagy protein Bnip3 along with basal autophagy markers in the mouse skeletal muscle [76]. Using the exercise-trained mice heterozygous for critical autophagy protein Atg6 the authors also showed that an attenuated increase of basal autophagy and mitophagy is associated with compromised endurance capacity. These findings demonstrate that EE-induced increase in basal autophagy and mitophagy is required for improved physical performance.

Together these findings elicit the complex interplay of the mitochondrial remodeling machinery involved in mitochondrial biogenesis, maintenance and clearance in the skeletal muscle. EE positively modulates each of these processes as part of muscle adaptation response towards enhanced metabolic and contractile capacity.

3.2. EE improves cellular aerobic fitness

In addition to enhancing mitochondrial biogenesis, EE also triggers an up-regulation of mitochondrial enzymes involved in the krebs cycle, electron transport chain and heme synthesis [77]. Additionally there is an up-regulation of enzymes involved in fatty acid oxidation and of the proteins mediating glucose entry (glucose transporter-4) into the cell [78,79]. Hence the relative contribution of fat to the muscle energy generation is greatly increased [80] along with a reduction in the rate of glycogen depletion [81] following EE. These adaptations improve the cellular capacity to generate ATP aerobically and the sensitivity of respiratory control in both type I and type II fibers during exercise [82]. EE alters many cellular properties resulting in enhanced aerobic capacity of the muscle fiber without changing the mean cross-sectional area. Both type I and type II muscle fibers show an increment in capillary density in response to EE [83,84]. An increase in myonuclear count per unit fiber length is reported in rats [85] and humans [86] in response to swimming and cycling, respectively. Since the fiber size is unaltered, the cytoplasmic volume per myonucleus (myonuclear domain) decreases optimizing the internuclear cooperation in metabolically active fibers. However, some studies report no increase in myonuclei content with EE [85,87]. In contrast, the amplification of muscle satellite cell

pool is a consistent finding in mouse [88], rat [89] and humans (Table 1) [87]. Hence, the muscle mass regulation and the expansion of satellite cells pool appear to work independently (at least partly) in EE. These findings indicate that the EE increases the muscle oxidative and regenerative potential by increasing the proportion of slow myosin, mitochondria, capillaries, nuclei and the satellite cells pool. These adaptations allow the athlete to exercise long for a given intensity or increase the intensity for a given exercise time in an energy efficient way. This is reflected by an improvement in all four parameters of aerobic fitness with EE, the maximal oxygen uptake ($\text{VO}_{2\text{ max}}$), exercise economy, the lactate/ventilatory threshold and the oxygen uptake kinetics [90–92].

4. Resistance exercise

Resistant exercise (RE) involves low-frequency, high intensity contractions against an external resistance with the aim of increasing the muscle bulk. Force production is high and in general $\approx 80\%$ of maximum power output is considered optimal [93,94]. Typical activities under this module include body building and weight lifting.

4.1. RE induces muscle fiber hypertrophy

It is generally believed that the hypertrophy of existing muscle fibers following RE is predominantly, if not entirely due to muscle hypertrophy from the addition of new sarcomeres and myofibrils in parallel [95] alongside increases in the amount of contractile motor proteins actin and myosin [96]. Growth and proliferation of individual myofibrils has been reported in muscle fiber hypertrophy due to resistance training [97]. These authors reported no change in myofibrillar density indicating an addition of myofibrils to the periphery of the muscle fiber. This is consistent with the finding of incorporation of new proteins at the periphery of muscle fiber. The increase in contractile proteins is mirrored by an increase in number of myosin cross-bridges resulting in concomitant increase in muscle fiber diameter and force-generating capacity. In addition to contractile proteins, mixed, mitochondrial and cytosolic proteins synthesis rates also increase [98,99] in response to bouts of RE. These responses are dose dependent and increase with increasing exercise intensity [99]. Further, RE also induces an increase, albeit proportionally smaller one in muscle protein degradation rate [100]. The rate of protein synthesis should supersede degradation for an extended period of time before hypertrophic response is evident [101,102].

RE leads to the preferential hypertrophy of type II fibers [103,104]. Indeed, type II fibers show greater plasticity in size when compared to type I fibers, in response to various mechanical stimuli. They hypertrophy and atrophy more rapidly in response to exercise and detraining. This phenomenon is evident in short-term (6–10 weeks) RE which only lead to type II fibers hypertrophy [105–107]. The longer studies, on the other hand report an increase in both type I and type II fibers area [108,109]. In a given muscle, type II fibers have slightly larger cross-sectional area [110] and specific force [111] than type I fibers. Hence the proportion of type II fibers is positively correlated with hypertrophy [112] and strength [113] in response to 'RE'.

In contrast to EE, RE has smaller effect on mitochondrial volume and PGC-1 α protein content. However, subcellular localization rather than total muscle PGC-1 α content might be a better indicator of PGC-1 α activity [114,115]. Immunofluorescence imaging shows that the PGC-1 α is localized both in the nucleus and the cytoplasm. The subcellular distribution predominantly shifts to the nucleus during early response to the oxidative stress [114]. Indeed, nuclear PGC-1 α protein is shown to preferentially

increase in response to low volume high intensity exercise [116] so that more PGC-1 α is present in nucleus for the initial adoptive response to training. Further, PGC-1 α 4, an isoform of PGC-1 α , is specifically expressed in response to RE in humans and regulates hypertrophic phenotype by inducing IGF-1 and repressing myostatin [51]. It is also shown to prevent skeletal muscle atrophy due to cancer-cachexia in mice [51]. However the precise role of PGC-1 α 4 in regulating skeletal muscle hypertrophy is controversial as it was not linked with chronic overload associated hypertrophy in synergistic-ablation mouse model [117]. The authors found no increase in p-Akt^{T308} in the synergistically ablated mice muscle and argue that this mode of hypertrophy is P13K-independent and hence might not involve PGC-1 α 4.

4.2. RE increases satellite cells and myonuclei count

A growing body of literature suggests that the number of satellite cells increases in response to short- [118–120] and long-term [121–123] resistance training (Table 1). This increase seems to occur in a fiber type-specific manner, and is mainly observed in type II fibers [124,125], despite the fact that the type I fibers contain a higher number of baseline satellite cells population [125,126] than type II fibers. The increment in satellite cells population of type II fibers is correlated with robust hypertrophy of these fibers in response to RE. Typically, a satellite cell undergoes mitosis and donates one of the daughter cells as a true myonucleus to the muscle fiber [127]. The new myonucleus, while in a post-mitotic stage, begins to produce muscle specific proteins that increase muscle size [128]. The increase in myonuclei counts, although a consistent finding in most hypertrophies [129,130] generally lags behind increase in fiber volume [131]. These findings support the notion of myonuclear ‘ceiling’ whereby existing myonuclei can support initially hypertrophy via enhancing protein synthesis [102,131] without the need for further myonuclear accretion. Once a certain ‘threshold’ for transcriptional activity is reached, further nuclei via satellite cells divisions are required to support hypertrophy [132–134]. Further, the ceiling limit differs for hypertrophy with and without functional competence where increase in size parallels or lags behind increase in force [135]. Despite these findings, the role of satellite cells in overload-induced muscle hypertrophy is still debated [136] and hypertrophy without satellite cells incorporation is reported in rodent models of satellite cells depletion [137,138]. For instance, ablation of >90% of satellite cells in mature mice skeletal muscle did not blunt the overload-induced hypertrophy in plantaris muscle [138]. However $\approx$ 2 fold increase in muscle weight within a short span of time may not adequately represent human adaptation [139]. Further, in humans a huge range of inter-subject variability is reported in hypertrophic response, and is attributed to variable sensitivity of mechanotransduction machinery to mobilize satellite cells [121,128,140].

4.3. RE enhances muscle net protein synthesis

Muscle hypertrophy is driven by net protein accretion with protein synthesis superseding degradation [102,141]. The acute response is intensity-dependent and is minimal at intensity <40% of one-repetition maximum (1-RM) but is 2–3 fold higher when muscle is activated at >60% of 1-RM intensity [99]. This suggests that the acute bouts of RE generate an anabolic response in a dose-dependent manner. The magnitude of response gets progressively smaller with time and is blunted by 3–6 weeks [142], reflecting desensitization of muscle anabolic machinery with repeated RE [143]. Not all the muscle proteins fractions respond to RE stimulus. Myofibrillar proteins show a preferential increase in synthesis rate in the individuals trained for RE [144–146] along with smaller

increase in protein breakdown [147]. However, this distinct response is blunted in untrained individuals where RE stimulus shows an additional increase in mitochondrial protein synthesis [146]. This is evident by an up regulation of mitochondrial enzymes involved in citric acid cycle (citrate synthase; [148,149]) and glucose phosphorylation in response to RE in untrained individuals. This is attributed to ‘unfamiliarity’ of exercise stimulus with larger disturbance of muscle homeostasis.

Muscle protein synthesis during hypertrophic response is induced through mechanical strain on the sarcolemma which induces signaling pathways through focal adhesion kinase (FAK), an attachment complex protein [150]. Activation of FAK induces mTOR which drives the mechanosensory pathways by activating P70^{S6K}, causing protein accretion (Fig. 2). This hypertrophic response is attenuated by the mTOR inhibitor, rapamycin [151] demonstrating the critical role of mTOR in muscle hypertrophy. mTOR enhances translational capacity and efficiency of muscle proteins leading to muscle fiber enlargement. These effects are primarily achieved via regulation of multiprotein complex mTORC1 [152] which relays signaling to P70^{S6K} and 4E-BP1. These effects are fiber-type specific and are more pronounced in muscles containing larger proportion of type II fibers [153,154] as against muscles primarily containing type I fibers [154]. Indeed, mTOR phosphorylation preferentially increases in type II fibers for up to several hours after a bout of RE [154]. Further, RE results in a greater phosphorylation of S6K1 in type II fibers compared to type I fibers [155] in alignment with preferential type II fibers hypertrophy. The subcellular localization of these two proteins also plays an important role in protein synthesis. Both the activation and phosphorylation of P70^{S6K} and 4E-BP1 have been correlated with nuclear mTOR, as nuclear import of mTOR enhances and export attenuates the phosphorylation/activation [156]. Similarly, using immunofluorescence staining p-S6K1 has primarily been found to localize in nucleus in pre-exercised status and in cytosol after the RE [155]. In accordance with the fiber type specific response these localizations are more pronounced in type II compared to type I fibers.

5. Eccentric exercise, injury and regeneration

Exercise induced injuries are common to the muscle and mostly involve eccentric or lengthening contraction. Such contractions, if not performed carefully, can cause excessive sarcomere strain leading to cellular membranes disruption and myofibrillar damage [157]. Type II fibers are more susceptible to such damage compared to type I fibers in humans [158], rabbit [159] and rodents [160]. This is attributed to ultrastructural and metabolic differences between two fiber types and is discussed in detail later in this review.

5.1. Ultrastructural changes in muscle fiber injury

Most of the eccentric-exercise induced damage at the sarcomere affects Z-disk which are the sarcomeric anchors for muscle structural integrity [161]. Thus in a mild form of injury, Z-disks appear wavy (i.e., Z-disk ‘streaming’) with slight myofibrillar disarray. In more severe cases, Z-disks disruption, thickening, focal losses and displacement into adjacent sarcomere are reported [162]. These changes are accompanied by disruption of other cytoskeletal elements especially desmin. Desmin is an intermediate filament that integrates the sarcolemma, Z-disk and nuclear membrane thus maintaining the longitudinal and transverse passive mechanical properties of the muscle fiber [163]. Eccentric contractions cause a segmental and rapid loss of desmin intermediate filaments that is not observed in inactivated muscle [164]. Further, the number of desmin-null fibers increases with repeated bouts of eccentric contractions, and the fibers lacking desmin

accumulate plasma fibronectin, indicating a loss of membrane integrity [164]. These changes are accompanied by disruption of the sarcomere [165], cytoskeleton elements [166] and sarcolemma [167], swollen mitochondria [158], displaced organelles [168] and impaired excitation–contraction coupling [169] resulting in reduced force generation [170].

5.2. ROS involvement in muscle injury

It is well established that the contracting skeletal muscle produces excess free radicals which are involved in regulating muscle adaption and function [171]. An augmented production of these free radicals during vigorous exercise produces oxidative stress.

ROS trigger pathways that regulate muscle adaptive changes to meet the physiological demand during exercise [172]. Indeed, moderate increase in ROS production in response to exercise stimulus is necessary for the remodeling of skeletal muscle [171]. However the effect of increased level of ROS on muscle structural integrity during strenuous exercise is still a subject of debate. Many studies report ROS involvement in exercise-mediated muscle injuries [173–175]. Increased level of ROS is detected following eccentric exercise using electron paramagnetic resonance spectroscopy [176] and indirect markers such as lipid peroxidation and protein oxidation [177]. It is postulated that the increased level of ROS may contribute to muscle injury as antioxidant supplements slightly attenuate the degree of delayed onset muscle soreness [174,178]. However, the ultrastructural damage due to eccentric exercise described above is not attenuated by anti-oxidants [179]. Further, ROS formation following eccentric exercise occurs after the maximal peak of muscle soreness has occurred [175]. Thus the precise relation between ROS and muscle structural damage following eccentric exercise remains unsubstantiated.

In contrast to the effects on muscle structure, the ROS mediated modulation of muscle force production is well characterized. Hence a modest increase in ROS in an unfatigued muscle positively modulates force production [180,181] as evident by antioxidant-mediated depression of force production [182,183]. However, when the ROS production crosses a certain threshold, a depression of force production and increased fatigue occur in skeletal muscle [184]. Conversely in such settings, antioxidant supplementation enhances the force production [185]. The modulation of force and fatigue properties by ROS and NO is achieved through numerous mechanisms including but not restricted to SR calcium regulation, myofilament structure and calcium sensitivity and mitochondrial ATP production [186,187]. Based on these findings a model was proposed where an optimal redox state is required for maximal isometric force production while a deviation from optimal redox state results in force depression [180].

Many signaling pathways are regulated by exercise-induced ROS generation leading to muscle remodeling. Among those, nuclear factor κ B (NF κ B), mitogen activated protein kinase (MAPK) and PGC-1 α play critical roles in the exercise physiology. Contractile activity triggers these pathways via increased H₂O₂ production. These pathways are involved in a wide variety of biological functions such as antioxidant defense, mitochondrial biogenesis, glucose transport and inflammation. A detailed description of these transcription factors is beyond the scope of this review and therefore, only a brief summary is presented. Readers are referred to several excellent reviews on this topic for detailed description [9,188–190].

NF κ B is a dimeric transcription factor in the sarcoplasm which is bound to its inhibitory subunit I κ B in its inactive form. Phosphorylation of I κ B releases P50/P65 subunits from NF κ B which then migrate to nucleus and bind to NF κ B binding sites of the promoter regions of a variety of target genes. NF κ B activation is redox sensitive and is blunted in response to reduced oxidative

stress in exercising rats [191]. The peak DNA binding activity is reported 2 h after I κ B phosphorylation and acute exercise stimulus [192] and is mainly restricted to oxidative fibers with higher mitochondrial contents [193]. As a transcription factor, NF κ B regulates expression of over 150 genes including those encoding apoptosis, oxidative defense, disuse atrophy and redox status [194]. With regards to injury, NF κ B is tipped to facilitate post-exercise regenerative potential in the damaged tissues by inducing acute phase proteins and proinflammatory genes [190]. Indeed, increased inflammation and protein turnover are prerequisites for exercise-induced hypertrophic and regenerative response [195].

The MAPK constitute a family of serine/threonine kinases that control multitude of cellular processes including muscle repair and growth. The pathways are activated by both extracellular (inflammatory cytokines, growth factors) and intracellular (ROS) activators. MAPK pathways are activated during exercise and regulate various muscle responses to exercise including hypertrophy, upregulation of antioxidant enzymes and fiber type transitions. These effects are severely hampered by treatment with xanthine oxidase inhibitors eliciting redox sensitivity of MAPK [191]. The physiological relevance of MAPK cascades in tissue repair process is elicited by their activation in the immediate vicinity of injury site in the human skeletal muscle [196]. Indeed, dysfunctional MAPK signaling via deletion of its upstream regulators play an important role in regenerative myogenesis and in the pathogenesis of muscular dystrophies [197].

PGC-1 α is considered a redox signaling pathway because ROS generated by exercise promote its expression along with its target genes [198]. Its role in exercise and muscle plasticity is discussed elsewhere in this review.

Many studies report ROS involvement in exercise-mediated muscle injuries [173–175]. Increased level of ROS is detected following eccentric exercise using electron paramagnetic resonance spectroscopy [176] and indirect markers such as lipid peroxidation and protein oxidation [177]. It is postulated that the increased level of ROS may contribute to muscle injury as antioxidant supplements slightly attenuate the degree of delayed onset muscle soreness [174,178]. However, the ultrastructural damage due to eccentric exercise described above is not attenuated by anti-oxidants [179].

Further, ROS formation following eccentric exercise occurs after the maximal peak of muscle soreness has occurred [175]. Thus the precise relation between ROS and muscle damage following eccentric exercise remains unsubstantiated.

5.3. Why are fast-twitch fibers more susceptible to eccentric exercise-induced damage?

Preferential damage to fast-twitch type II fibers due to eccentric exercise was first reported more than three decades ago [199] in human vastus lateralis muscles with a damage ratio of 3:1 between type II and type I fibers. These findings were subsequently confirmed in human gastrocnemius and biceps brachii muscles too [200]. Many animals and humans studies confirm these findings. For example, human quadriceps muscle show preferential type II fibers damage 72 h after plyometric exercise [158]. Similar findings have been reported in flexor muscles of elbow and foot [200]. Previously, this was attributed to early fatigue and prolongs rigor binding state of fast-twitch fibers causing mechanical damage when rigor fibers are stretched [201]. However subsequent studies found that the fatigued fibers are less susceptible to eccentric exercise induced damage [202,203].

It is now believed that the ultra-structural differences between the two fiber types account for higher vulnerability to damage of type II fibers (Fig. 3).

These cytoskeletal elements maintain structural integrity of muscle fiber during repeated mechanical stress and thus prevent it

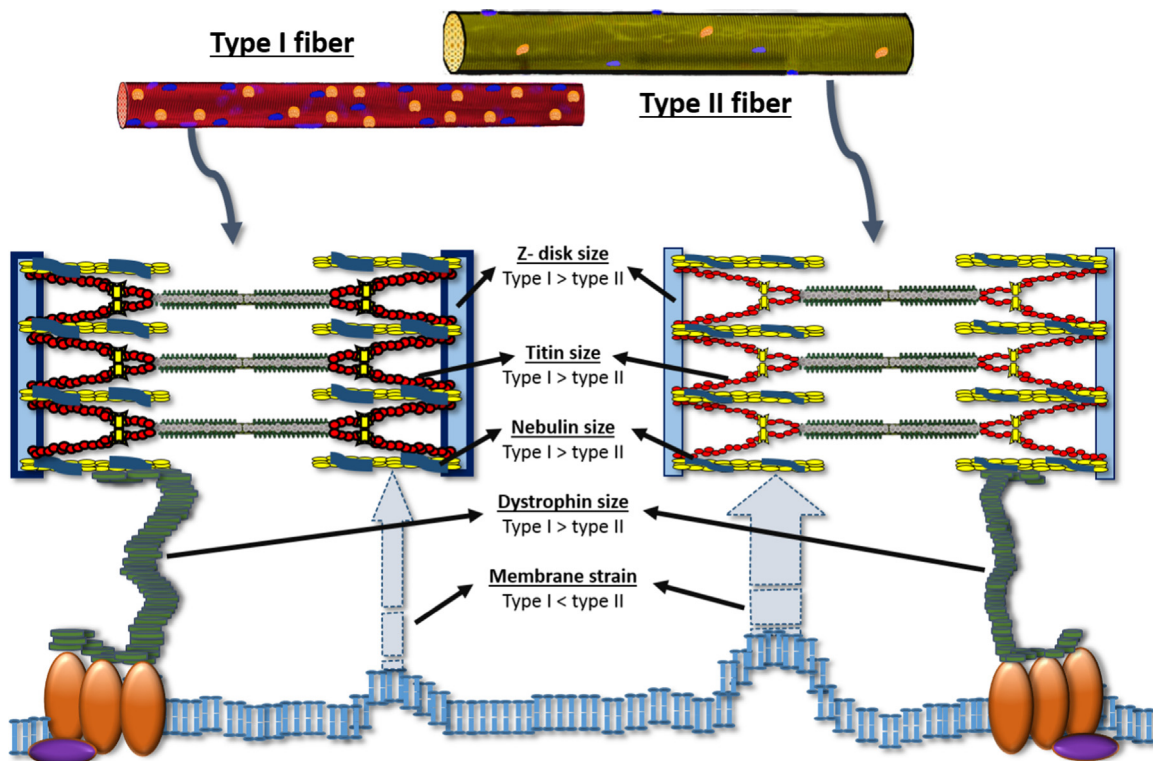


Fig. 3. Ultrastructural differences between type I and type II fibers related to exercise-induced damage. Type II fibers are more susceptible to injury due to smaller Z-disk, titin, nebulin and dystrophin as well as higher strain on the sarcolemma.

from injury. Type II fibers have narrow Z-disks [158] compared to type I fibers which reflect fewer attachments for sarcomeric filaments [204]. Type II fibers also express smaller isoforms of nebulin and myomesin which play important roles in sarcomeric assembly [205]. Further, adherent protein titin which spans half the sarcomere and regulates sarcomeric integrity is distinctly expressed in two fiber types. Type II fibers express low molecular weight, less elastic isoform than type I fibers [206] resulting in reduced resistance to strain. The structural vulnerability of cytoskeleton and sarcomere is compounded by the higher vulnerability to degradation of desmin and troponin T in type II fibers [207]. When the sarcomere contract, the longitudinal force vector is transmitted to the sarcolemma, hence sarcolemma integrity is pivotal to muscle fiber structural integrity. Dystrophin, a sarcolemma associated cytoskeletal protein has two fold higher content in type I compared to type II fibers [208]. Thus the type II fibers have less ability to absorb strain on the outer membrane. These fibers also have slightly higher specific force than type I fibers [111]. Thus, more force is exerted per unit surface area of sarcolemma in type II fibers. Compounded by structural weakness, this makes type II fibers inherently vulnerable to strain-induced damage. This vulnerability has been studied in a fiber type-specific manner where single fibers were subjected to standardized eccentric contractions and the force generating capacity was studied before and after contractions [209]. Force generation and force deficit before and after eccentric contractions were in harmony so that the stronger fibers showed greater force loss following eccentric contraction. In an isoform-specific manner, fibers expressing stronger and faster isoforms showed more damage.

In addition to structural proteins, type II fibers also express decreased level of 'stress proteins' including heat shock proteins that protect the fiber from mechanical stress during eccentric contractions and also accelerate recover after the stress stimulus is gone [210]. These fibers also have less regenerative capacity than type I fibers due to smaller number of nuclei and satellite cells.

6. Regeneration

The remarkable ability of skeletal muscle to induce repair and regeneration following injury is well documented [211,212]. Activation and proliferation of muscle satellite cells are key components of this process. This is evident by impaired regeneration after local depletion of satellite cells following exercise-induced muscle injury [213]. The muscle regenerative capacity is rescued by the transplantation of Pax7⁺ satellite cells showing that the satellite cells are essential for acute injury related regeneration. Satellite cells are associated with all fiber types albeit with unequal distribution. For instance, slow-twitch muscles express two- to three-folds higher satellite cells density when compared to fast-twitch muscles [214,215]. These findings are reported in single muscle fibers too [216]. Further, phenotype of satellite cells from adult muscle fibers is similar to their fiber of origin and the fiber they tend to form [217,218]. This distinction also applies to various fiber types within the same muscles and points towards differential programming of different satellite cells.

Upon injury, quiescent satellite cells get activated and start proliferating, and are now referred to as adult myoblasts. The activation of adult myoblasts is characterized by up regulation of myogenic regulatory factors (MRFs), *Myf5* and *MyoD* while fusion is mediated by cell adhesion molecule M-cadherin although N-cadherin and R-cadherin can compensate for its deficiency. A detailed description of these molecular regulators of muscle regeneration can be found elsewhere [211]. Early stages of regeneration are characterized by expression of developmental myosin isoforms [219] followed by a switch to adult isoforms under the control of nerve activity. The type of default switch to adult myosin isoform type depends on the innervation status of the muscle. Thus the regenerating rat soleus muscle acquires adult slow-twitch profile when innervation is intact and fast-twitch profile when denervated [220,221]. Further, low frequency stimulations of the denervated soleus muscle capitulates the slow

twitch profile showing that the impulse frequency of the motor neurons dictates the muscle fiber type profile during regeneration [222].

7. Signaling pathways regulating fiber type switching

A sophisticated signaling network controls exercise-induced fiber type profiling within skeletal muscle. This network senses intracellular calcium concentration and metabolic stress to activate various transcription factors involved in fiber type transition. Indeed the type and activity of contraction determines intracellular calcium concentration [223] and different signaling pathways are activated by various intracellular calcium concentrations promoting fast or slow phenotype. The regulation is antithetical so that the promotion of fast phenotype causes repression of slow phenotype and vice versa. The slow-twitch phenotype is predominantly promoted by the activation of calcineurin-dependent nuclear factor of activated T-cells (NFAT) transcription factors [224] as blockage of this pathway blunts the slow phenotype promoting activity in the muscle [29]. Further, Ca^{2+} /Calmodulin-dependent protein kinase (CaMK) which decodes frequency dependent information [225] also induces exercise-induced slow phenotype by activating MEF2 transcription factors. Conversely, transgenic mice with a hyperactive form of MEF2 show an increased proportion of slow-twitch fibers and enhanced running endurance when compared to wild-type counterparts [226]. PGC-1 α and PPAR β also participate in regulating slow oxidative phenotype [227] in response to EE, although PGC-1 α is probably more involved in maintaining rather than inducing a slow-twitch phenotype [228].

Relatively less is known about the signaling pathways regulating switch from slow to fast phenotype during intense exercise. Skeletal muscle reprogramming from slow oxidative to fast glycolytic phenotype involves members of the Six1/Eya1 complex that are found in higher concentration in the nuclei of fast gastrocnemius compared to slow-switch soleus muscle. Moreover they induce the transition from oxidative to glycolytic phenotype when overexpressed in soleus muscle [229]. The muscle glycolytic enzymes activity during intense exercise is regulated by hypoxia inducible factor 1- α (HIF-1 α) as shown by altered exercise endurance with deletion of HIF-1 α [230]. The expression of myogenesis regulatory protein, MyoD is different in fast- and slow-twitch muscle fibers and thus influences the fiber type composition of muscle [231]. Indeed over-expression of MyoD is associated with increase number of fast-twitch glycolytic fibers in predominantly fast- [232] as well as slow-twitch muscles [233]. These findings are consistent with the hypothesis that the MyoD is required for fast glycolytic fibers formation.

8. Conclusion

Skeletal muscle is a very dynamic tissue which is composed of individual muscle fibers with a dynamic range of chemical, biomechanical and physiological properties. The presence of diverse fiber types with distinct ranges of adaptability reflects muscle plasticity to various external stimuli including exercise training. Endurance and resistant trainings on account of different intensity and duration reflect specific patterns of mechanical strain which prepare muscle to various functional tasks. This is achieved by modulating muscle fiber phenotype in a predetermined range of adaptation as well as fiber type switching to desired phenotype. Another area of attention in this review is exercise-induced injuries and the mechanisms of susceptibility of type II fibers to damage. Signaling pathways controlling fiber type diversification

and maintenance during muscle adaptation process are briefly summarized. Promoting muscle fiber type switching and modulating specific fiber types can help to counter various muscle wasting conditions. This would help to build therapeutic strategies for targeted exercise interventions based on mode, duration and intensity of exercise.

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